Data Generation Philosophy

SYNTHETIC TELEMETRY ENGINEERING FOR SYNCOS

Simulating SyncOS operational telemetry presented a core challenge: generating data that reflects physical reality without hardcoding the exact bottlenecks I needed to discover.

Phase 1: The Baseline Fallacy

Initial datasets relied on uniform randomness. Task durations had arbitrary boundaries, and spatial coordinates were equally probable across the facility map.

> **The limitation of pure randomness:**
> It fails to test spatial and temporal hypotheses. Traffic blockages spread uniformly, and battery levels fluctuated randomly. This made identifying root causes like staging deadlocks impossible.

Phase 2: Procedural Reality

I transitioned to procedural synthetic generation. Real-world operations follow structural logic, not uniform randomness.

Instead of artificially inserting traffic jams, I modeled the physical constraints that cause them. By defining natural battery drain, shift-based task clustering, and zone traffic weighting, the dataset organically produced authentic bottlenecks, serving as a reliable proxy for a live production environment.

Core Simulation Techniques

Gaussian Task Durations

I replaced uniform bounds with normal (Gaussian) distributions. Modeling task times on bell curves mirrors the natural rhythm of physical machinery and human handovers.

Spatial Weighting

I assigned logical probabilities to facility zones. Skewing traffic generation toward the 'Dispatch Zone' accurately reflects real-world layout dynamics and physical congestion.

Chronological State Tracking

I implemented sequential hardware state trackers. Draining batteries based on movement and recharging them sequentially allowed me to measure true reactive charging logic.

Temporal Modifiers

I modulated task generation via time-of-day curves. Simulating higher task frequencies during daytime shifts created realistic queuing pressure to stress-test the system.

Impact on Hypothesis Testing

Procedural generation was critical to my analysis. It allowed me to validate Hypotheses 3 (Staging Deadlocks) and 4 (Reactive Charging) as natural consequences of structural constraints. Conversely, I confidently invalidated Hypothesis 2 (Shift Change Bottlenecks) because station handovers maintained a stable distribution despite temporal pressure.